AgroPlus

Soil & weather → grounded AI crop & profit advisor for Bangladeshi farmers

SE 801 · SOFTWARE PROJECT LAB–III

Md. Tuuhidur Rahman | Flutter + FastAPI + PostgreSQL

Flutter (Android/iOS) · FastAPI · PostgreSQL · XGBoost + Random Forest · OpenAI Function Calling · Open-Meteo · BAMIS · DAM · BADC Data · EN / বাং Localized

Soil-Based Recommendation

Tiered sampling → composite N/P/K/pH → ML-ranked top-3 crops, confidence-scored

XGBoost + Random Forest ensemble

Itemized Profit Engine

Seed + per-nutrient fertilizer + other cost, revenue and net profit, per acre

Seed + Urea/TSP/MoP/Gypsum breakdown

Double / Triple Cropping

Season-sequenced plan with automatic substitution when a season's best pick is blocked

No crop grown > 2× in a row

AI Farm Assistant

Tool-calling chat grounded only in the land's real data — never fabricates a number

6 grounding tools · persistent per-land thread

Live Weather & 7-Day Forecast

Current conditions plus daily high/low temperature and rainfall outlook per district

Open-Meteo, per-district coordinates

"Why This Crop?"

Compares the land's actual soil/weather values against each crop's typical range

One tap from crop detail → chat

Budget-Aware Warnings

Flags any crop whose estimated total cost exceeds the farmer's stated budget

Shown on list & detail screens

History & Land Management

Full CRUD on land profiles plus chronological recommendation history per land

Edit, delete, re-sample anytime

DATA FLOW

Soil Sample → ML Engine → Economics DB

Live Weather → Recommendation / Plan → AI Chat

BAMIS · DAM · BADC · Open-Meteo

DESIGN PRINCIPLES

- ✓ **Grounded, not generative** — every number the assistant states comes from a real tool call, not model knowledge.
- ✓ **Provenance over estimation** — yield/price/cost always traced to a named dataset.
- ✓ **Rotation-safe by default** — the same exclusion rule applies everywhere crops are ranked.
- ✓ **Component-based backend** — ML, economics, weather, and chat are independently replaceable services.
- ✓ **One codebase, two languages** — Flutter + full English/Bangla localization, agent replies in-language.

10 backend components · 8 database entities · 36 functional requirements · 2 languages

Auth · Land Profiles · Soil Sampling · ML Engine · Economics · Weather · AI Chat